Handout 5b: Taxes and Deadweight Loss

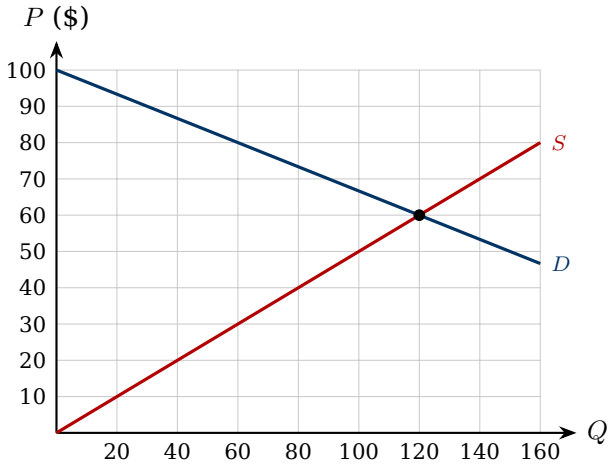
Mankiw Ch 7–8

A tax on kayak rentals

The daily market for kayak rentals on the Charles River (Handout 5, Problem 2) is

$$Q_d = 300 - 3P \qquad Q_s = 2P$$

Without a tax: \$60 a rental, 120 rentals a day, total surplus \$6,000. The city puts a **\$20 tax** on each rental, paid by the renter.



a) What price do renters pay, what do owners keep, and how many kayaks are rented? Who bears more of the tax, and why?

b) Shade consumer surplus, producer surplus, tax revenue and the deadweight loss on the grid. Compute each, and total surplus.

c) The city doubles the tax to \$40. Find the new quantity, revenue and deadweight loss. Did revenue double? By how much did the deadweight loss grow, and why?

d) Suppose instead there are exactly 120 kayaks, and owners rent them all at any price (supply is vertical at 120). Who bears the \$20 tax, and what is the deadweight loss?